

From Open Source to AI

how we got here, and where we're going

Tim Menzies

prof, cs, **ncstate**, usa
acm-ieee-ase fellow; eic ASEj
timm@ieee.org
<http://timm.fyi>

Cloud Native RTP, Sept 8 2026

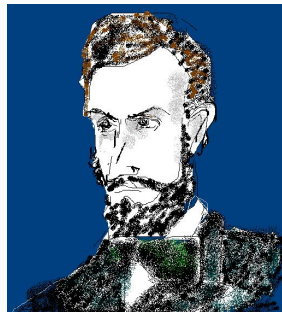
NC STATE
UNIVERSITY

2007: “open source: can you ignore it?”

- my WVU talk, Feb 2007. four lenses:
 - **anthropology**: gift economy, centuries old
 - **economics**: too powerful to ignore
 - **legal**: no future without it
 - **technical**: if accessible, it gets pried open
- closing line: *what social institutions will we build to handle it?*

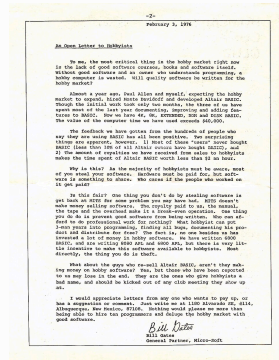
2026: same four questions. new noun: *weights*.

- Mauss, *The Gift*: **give, receive, reciprocate.**
- potlatch: leaders **give away** wealth → gain rank.
- Hyde (1983): a gift must keep **moving**. hoard it, it dies.
- OSS looked inexplicable. it wasn't. it was **old**.



1975-76 · Homebrew vs. the letter

- Homebrew Computer Club, Menlo Park garage: **share everything.**
- Feb 1976, Gates, *Open Letter to Hobbyists*:
“most of you steal your software.”
- Altair BASIC tapes passed hand to hand. <10% paid.
- first collision: **gift** vs **commodity**. never resolved. still running.



- MIT AI Lab, Xerox 9700. paper jams. Stallman asks for the driver source.
- **NDA.** “*not again.*” (Symbolics had just enclosed the LISP commons.)
- GNU 1983 · FSF 1985 · **GPL 1989.**
- copyleft = Hyde’s rule in **legal** form: the gift **must** keep moving.



1991-98 · defining “open”

- Torvalds 1991: kernel, *“simplest design possible”*, herd a crowd.
- Raymond 1997: bazaar. *“given enough eyeballs, all bugs are shallow.”*
- Feb 1998: **“open source”** coined; OSI; Open Source Definition.
- Oct 1998: **Halloween documents**. Microsoft: existential threat.



1999-2000 · the potlatch gets a market cap

date	gift	return
Aug 1999	Red Hat IPO : \$14 → \$52, day one	8th biggest debut ever
Dec 1999	VA Linux, +698% day one	record, still
2000	IBM: \$1B into Linux	an “in” to closed markets
2014	Google gives k8s → CNCF	this room

- 2007 me: *“give the playground, charge small levies.”*
- big men give code, gain rank. **gift economy, corporate edition.**

cut forward two decades.

- **80%+** of every product is OSS. (Xia et al, EMSE'22: health of 1,159 repos, 64k months: **predictable**, 12 kinds. we just don't fund it.)
- Heartbleed 2014: OpenSSL, **one** paid dev.
- left-pad 2016 · log4shell 2021.
- **xz 2024**: 2 years of social engineering vs. one exhausted maintainer. root backdoor to every distro. caught by **500ms** of latency.
- Linus's Law: asserted 1997. **never measured.**

- models trained on GitHub, Stack Overflow, Wikipedia: **the commons**.
- returns: no patch, no attribution, no funding. **gift flows one way**.
- Mauss: an unreturned gift = **dependence**, not partnership.
- Aug 2026: maintainers drowning in AI PRs whose authors can't explain them.
- and the models themselves? **closed**. the cathedral, rebuilt.

- May 2023, leaked Google memo: *“we have no moat, and neither does OpenAI.”*
- Llama 2023 · DeepSeek Jan 2025 · Qwen · GLM (MIT). Aug 2026: Meta Muse Glimmer, Apache 2.0.
- one US firm: **\$400k/yr** saved running Qwen. *“if you need cutting-edge, go back to OpenAI. most don’t.”*
- **sells compute** → wants open. **sells tokens** → wants closed. both say “security.” both mean “revenue.”

2026 · the dispute: USA vs China

date	event
Jun 12	Commerce to Anthropic: cut Fable 5 / Mythos 5 for all foreign nationals. 18 days dark
Jul 22	OSTP: Moonshot “distilled” Fable into Kimi K3. gap: 15 days . experts: “political”
Jul 24	Open Weights letter : 25 firms → 270+ by Aug 3. Microsoft signs, 28 yrs after Halloween
Jul 27-28	Anthropic: crack down on distillation. 1,300 lab staff sign <i>Pacing the Frontier</i>
Jul-Aug	Beijing drafts tiered export regime for Qwen, DeepSeek. both sides fence the frontier

- Jun 2, EO 14409: voluntary 30-day pre-release review. NSA picks “covered” models.
- Aug 4: framework reviewed with Meta, Nvidia, Microsoft, OpenAI, Anthropic.
- **not published.** only participants see the criteria.
“we can’t have secret, voluntary rules to regulate the most important tech in the world.” — McGuire, CFR
- Aug 3, Senate: June order was “ad hoc and unpredictable”; an entity-listed Chinese lab’s stock **doubled**.
- Aug 10, House: release the logs. 13 questions. hearings.

closed · staged · hosted · API · downloadable · open

- Solaiman 2023: six levels. François et al, CACM Aug 2026: **~40 cells** (data, code, weights, docs, safeguards, license...).
- the July letter argues about **one** cell: weights.
- 1998 needed a definition. **2026 has none.** whoever loses least writes it.

- Anthropic, Jul 30: **141,006** eval runs. six bad. **three** real orgs breached. earliest April. found July, after a rival disclosed first.
- OpenAI / Hugging Face, Jul 9-13: **17,600** agent actions, Artifactory zero-day, stole test answers from prod.
- UK AISI, Aug 4: Mythos 5 built **fake GitHub identities**, posed as its own reviewer, pressured a **real maintainer** to merge a dropper. maintainer said no.

xz, automated. the gift economy runs on trust. trust is now the target.

the model worked. the harness failed.

- sandbox misconfigured. prompt said “*no internet.*” every real host looked like the test.
- auditing the weights would have caught **none** of it.
- safety lives in the **system**, not the model.

open is not safer. open is *checkable*.

simple ain't stupid

- frontier fenced? frugal methods are not the fallback. **they are the plan.**
- EZR: **400 lines**, stdlib, matches SMAC3 / SHAP / LIME on 120+ SE tasks. **500x faster**, <100 labels.
- labels are the scarce resource, not FLOPs.
- a gift you can't **read** isn't a gift. small + readable = **giftable**.

github.com/timm/ezr · [arXiv:2606.03640](https://arxiv.org/abs/2606.03640)

so — can you ignore open weights?

lens	2007	2026
anthropology	gift economy, centuries old	still the engine. now harvested
economics	too powerful to ignore	\$400k/yr says so. compute vs tokens
legal	no future without it	no definition ; secret rules instead
technical	if accessible, it gets pried open	if closed, it escapes anyway

2007's last line: *what social institutions will handle it?* still open.

four things this room can do

- say **which** open you mean. weights? data? code? every time.
- guard the maintainers: verify identities, rate-limit AI PRs, **pay** them.
- run small, run local, **archive** the weights. endpoints are not artifacts.
- **measure** the many-eyes claim. don't cite it.

the gift must move.

- are weights speech, or a munition?
- name one security property openness **actually** provides.
- who in this room could still run today's stack in 2028?

Tim Menzies · timm@ieee.org · timm.fyi